

May Ahmed

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Skills

Programming	Python, Bash, R; object-oriented design, Git/GitHub
Cheminformatics	Molecular docking (Glide, GNINA); structural search (Foldseek); RDKit, PDB2PQR, Open Babel; UCSF ChimeraX
Scientific Computing	NumPy, SciPy, pandas, Polars, PyArrow, Biopython, MDAnalysis; Linux/SLURM, HPC
Machine Learning	scikit-learn, PyTorch; supervised learning, feature engineering

Research Experience

Graduate Student Researcher — Dr. Jacob D. Durrant

DEPARTMENT OF BIOLOGICAL SCIENCES, UNIVERSITY OF PITTSBURGH

Sep 2021 – Present

- Designed **LIGNOVA**, an open-source Python pipeline integrating PubChem bioactive ligands with GNINA docking to generate large-scale receptor-bound structural data for structure-based drug design ML training; **125M poses across 175,200 compounds and 4,180 protein structures** generated to date. Recently awarded a competitive **NSF ACCESS allocation (BIO260240)** to scale.
- Tested whether **consensus-based rescoring (ComBind)** across **2,713 PubChem ligands** can substitute for explicit water modeling in molecular docking (i.e., recovering the pose-accuracy benefits of hydration treatment while preserving the throughput needed for virtual screening); evaluated across **255 protein-ligand structures** under three hydration conditions. Manuscript in preparation.
- In collaboration with the O'Donnell lab, led a **structure-based discovery pipeline** to identify candidate **arginine-sensing proteins** in *Saccharomyces cerevisiae* using **Foldseek similarity search and affinity profiling**; generated hypotheses **experimentally validated** to show altered arginine homeostasis phenotypes.

Undergraduate Student Researcher — Dr. Eman Badr

UNIVERSITY OF SCIENCE AND TECHNOLOGY, ZEWAIL CITY

Jun 2020 – Apr 2021

- Trained **Random Forest regression models** (scikit-learn) to predict drug sensitivity from transcriptomic profiles in **AML patients**; identified genetic markers correlated with differential drug response.

Education

Department of Biological Sciences, University of Pittsburgh

PH.D. CANDIDATE IN MOLECULAR, CELLULAR, AND DEVELOPMENTAL BIOLOGY, DR. JACOB DURRANT

Expected graduation: 12/2027

Aug 2021 – Present

Research focus: **cheminformatics**, structure-based **drug design**, and scientific software development.

University of Science and Technology, Zewail City

B.S. IN COMPUTATIONAL BIOLOGY AND GENOMICS (BIOMEDICAL SCIENCE)

Aug 2017 – Jul 2021

Graduated **cum laude**; awarded a **90% merit-based scholarship** throughout all my academic studies.

Publications

Holland, M.*, **Ahmed, M.***; Young, J. M.; Drurey, J. R.; McFadyen, S.; Ostrowski, E. A.; Levin, T. C. Hypermutable hotspot enables the rapid evolution of self/non-self recognition genes in *Dictyostelium*. *PNAS* **2025**, 122(51). DOI: 10.1073/pnas.2520843122. (*co-first authors)

Kochnev, Y.; **Ahmed, M.**; Maldonado, A. M.; Durrant, J. D. MolModa: Accessible and secure molecular docking in a web browser. *Nucleic Acids Research* **2024**, 52(W1), W498–W506. DOI: 10.1093/nar/gkae406.

Ahmed, M.; Maldonado, A. M.; Durrant, J. D. From byte to bench to bedside: Molecular dynamics simulations and drug discovery. *BMC Biology* **2023**, 21(1), 299. DOI: 10.1186/s12915-023-01791-z.

Conferences

2024	Poster Presenter — Beyond the Blueprint: A Novel Database for Innovative Drug Design and Discovery , Gordon Research Conference & Seminar on Computational Chemistry	Portland, ME
2023	Poster Presenter — Augmenting Protein-Ligand Complex Databases with PubChem for Enhanced Drug Discovery , Gordon Research Conference & Seminar on Computer-Aided Drug Design	West Dover, VT

Teaching & Mentorship Experience

Sum'24	Undergraduate Research Mentor , TECBIO REU; Co-mentored an AlphaFold-based drug design project	Pittsburgh, PA
Sp'24-	Graduate Teaching Assistant & Fellow (Biochemistry) , University of Pittsburgh; mentored 15	Pittsburgh, PA
Fall'25	undergraduate TAs; nominated for the Elizabeth Baranger Excellence in Teaching Award (both semesters).	
Fall'20	Undergraduate Teaching Assistant (Python) , Zewail City; helped students in writing and debugging scripts	Giza, Egypt